

Indian Institute Of Technology Gandhinagar



Assignment Report

Student Name: Bilavath Prem Singh

Roll Number: 25210098

Branch: Integrated Circuit Design And Technology (ICDT)

Department: Electrical Engineering

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Submitted to: Madhav Pathak

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Task 2: Designing a Current-Starved Voltage Controlled Oscillator (8 pts)

1. Design a Non-Linear Current-Starved VCO (refer to Fig. 1) and report the frequency of oscillation.
2. Vary the input voltage (control voltage) of the circuit and observe how the oscillation frequency changes with the control voltage. Vary the control voltage from 0.5 V to 1.8 V with a step of 0.2 V and report the oscillation frequency at each step.
3. Design a Linear Current-Starved VCO (refer to Fig. 2) for the oscillation frequency according to the table provided.
4. Repeat Part 2 for the Linear Current-Starved VCO as well.
5. Compare the two Current-Starved VCOs.

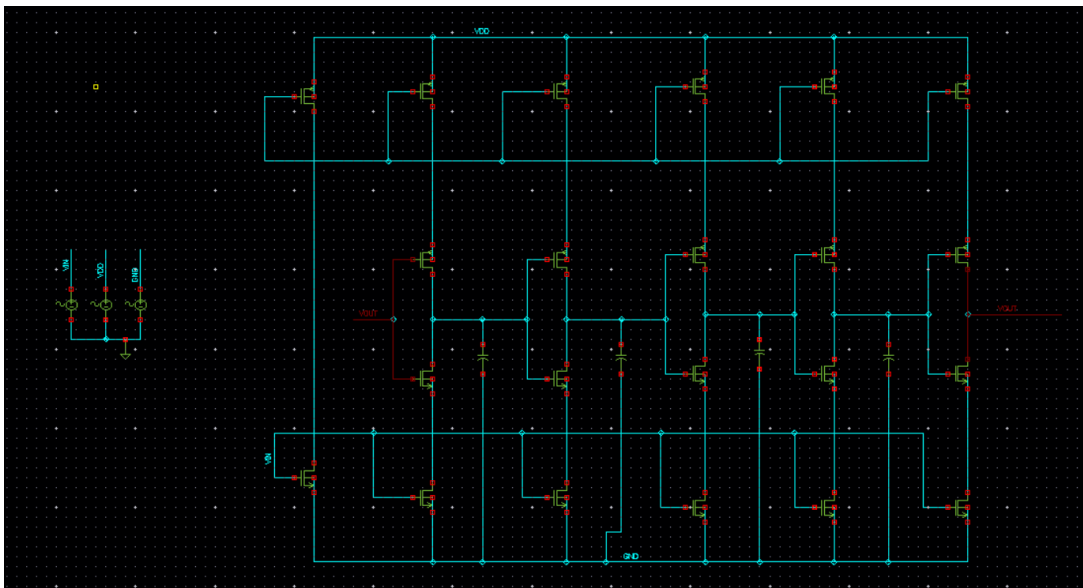


Figure 1: Circuit for Non-Linear Current Starved VCO

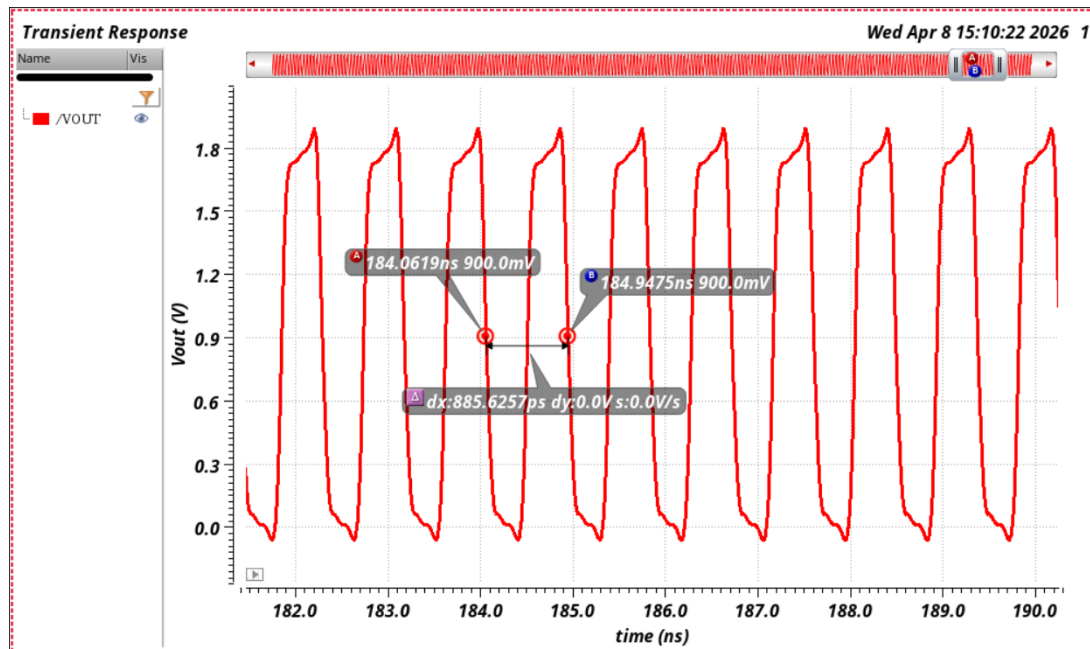


Figure 2: Frequency for Non-Linear Current Starved VCO

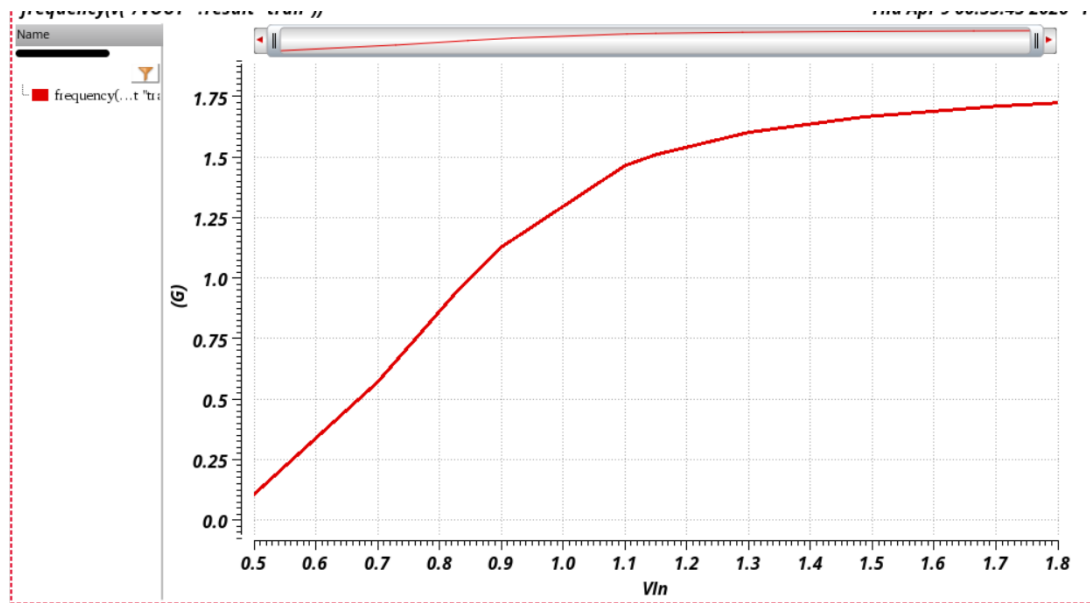


Figure 3: frequency vs input voltage

Control Voltage V_{ctrl} (V)	Oscillation Frequency (GHz)
0.5	0.1058
0.7	0.570
0.9	1.12765
1.1	1.46466
1.3	1.60205
1.5	1.669
1.7	1.70991
1.8	1.72386

Table 1: Control Voltage vs Oscillation Frequency for Non linear VCO

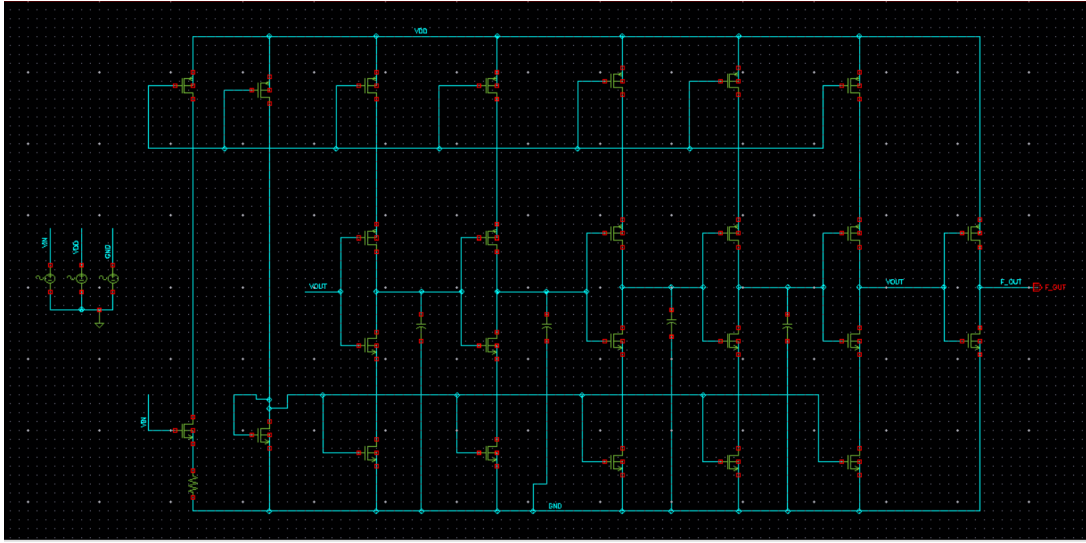


Figure 4: Circuit for Linear Current Starved VCO

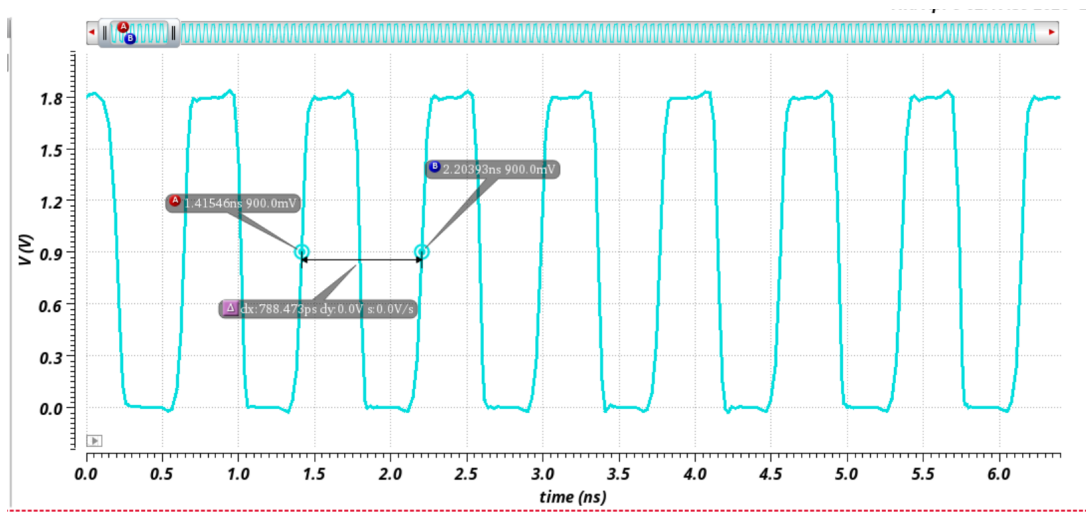


Figure 5: Circuit for Non-Linear Current Starved VCO

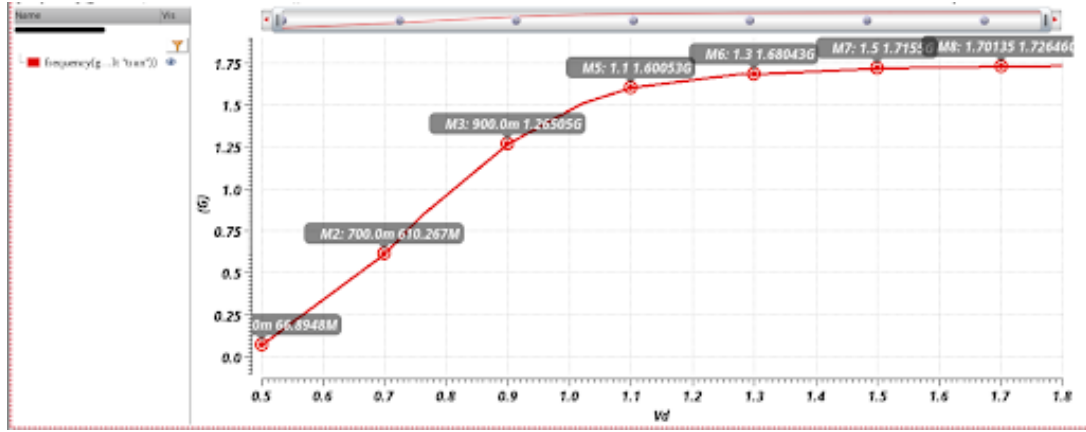


Figure 6: Control Voltage vs Oscillation Frequency for linear VCO

Control Voltage V_{ctrl} (V)	Oscillation Frequency (GHz)
0.5	0.066
0.7	0.61
0.9	1.265
1.1	1.600
1.3	1.680
1.5	1.7155
1.7	1.7264
1.8	1.7308

Table 2: Control Voltage vs Oscillation Frequency for linear VCO

Voltage (V)	Frequency (GHz) - Non-Linear VCO	Frequency (GHz) - Linear VCO
0.5	0.1058	0.066
0.7	0.570	0.61
0.9	1.12765	1.265
1.1	1.46466	1.600
1.3	1.60205	1.680
1.5	1.669	1.7155
1.7	1.70991	1.7264
1.8	1.72386	1.7308

Table 3: Control Voltage vs Oscillation Frequency for Linear and Non-Linear VCO